

## Project Design Phase-II

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|---------------|-----------------|
| Date          | 13-07-2026      |
| Team ID       | SWTID-2026-7371 |
| Project Name  | House Rent      |
| Maximum Marks | 4 Marks         |

### Functional Requirements:

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                               |
|--------|-------------------------------|--------------------------------------------------------------------------------------------------|
| FR-1   | User Registration             | Registration through Registration Form<br>Email and Password Validation<br>User Account Creation |
| FR-2   | User Login                    | Login using Email and Password<br>JWT Authentication<br>Secure User Session                      |
| FR-3   | Property Management           | View Available Properties<br>Search Properties by Title or Location<br>View Property Details     |
| FR-4   | Booking Management            | Book Rental Property<br>View My Bookings<br>Booking Confirmation                                 |
| FR-5   | Admin Management              | Add New Property<br>Update Property Details<br>Delete Property                                   |
| FR-6   | Database Management           | Store User Details<br>Store Property Information<br>Store Booking Records in MongoDB Atlas       |

### Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                              |
|--------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Usability                         | The application provides a simple, responsive, and user-friendly interface for easy property browsing and booking.                              |
| FR-2   | Security                          | User passwords are encrypted using <b>bcrypt.js</b> , and secure authentication is implemented using <b>JWT</b> .                               |
| FR-3   | Reliability                       | The system provides reliable property and booking management with secure database operations using MongoDB Atlas.                               |
| FR-4   | Performance                       | The application quickly loads property data and processes booking requests with minimal response time.                                          |
| FR-5   | Availability                      | The system is available whenever the frontend and backend servers are running, allowing users to access the application continuously.           |
| FR-6   | Scalability                       | The MERN Stack architecture supports future enhancements such as online payments, user profile management, notifications, and cloud deployment. |